

Advanced (Habanero)

- Q1.** What does the box in a box-and-whisker plot represent?
(A) The entire data set
(B) The median and quartiles
(C) The mean and standard deviation
(D) The range of the data
(E) The mode of the data
- Q2.** Which of the following is Q_3 in a box-and-whisker plot?
(A) The lower quartile
(B) The median
(C) The upper quartile
(D) The interquartile range
(E) The range of the data
- Q3.** What is the interquartile range (IQR) of the data set $\{1, 3, 5, 7, 9\}$?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q4.** Which data point is an outlier in the set $\{1, 2, 3, 4, 5, 100\}$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 100
- Q5.** What does the whisker in a box-and-whisker plot represent?
(A) The median and quartiles
(B) The range of the data
(C) The interquartile range
(D) The minimum and maximum values
(E) The mean and standard deviation
- Q6.** If $Q_1 = 2$ and $Q_3 = 6$, what is the IQR ?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q7.** Which of the following box-and-whisker plots represents the data set $\{1, 2, 3, 4, 5\}$?
(A) A plot with $Q_1 = 1$, $Q_3 = 5$
(B) A plot with $Q_1 = 2$, $Q_3 = 4$
(C) A plot with $Q_1 = 1$, $Q_3 = 3$
(D) A plot with $Q_1 = 3$, $Q_3 = 5$
(E) A plot with $Q_1 = 2$, $Q_3 = 3$
- Q8.** If the IQR is 4, and $Q_1 = 2$, what is Q_3 ?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12

Open Ended Questions (FRQ)

- Q9.** Construct a box-and-whisker plot for the data set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and identify the IQR .
- Q10.** A data set has $Q_1 = 5$, $Q_3 = 15$, and a minimum value of 2. Sketch the box-and-whisker plot and identify any outliers if the maximum value is 100.

Chef's Tips

- Tip 1: Ensure students understand the concept of quartiles and interquartile range.
- Tip 2: Remind students to carefully examine the data set for any outliers.
- Tip 3: Encourage students to use graph paper when constructing box-and-whisker plots.